

Final Project

Part 1: User Research & Design

The goal of this part of the assignment is to help you understand and empathize with your users and to understand the market and competitors before diving into development. This process ensures that you build a mobile application that people actually need and will use. According to startup failure analysis, the top reason products fail is “no market need”; not because of bugs, pricing, or marketing. To avoid this, you must empathize with users and understand the market and competitors.

Design is an iterative process, and it is usually recommended that you understand user experiences, specify their needs, design solutions, and evaluate them repeatedly. This phase is crucial before writing any code.

What You’ll Do

Preliminary Research and Literature Review

Before you speak to a single user, you must ground your problem in evidence. This means conducting a brief literature review to understand the existing research, data, and reports related to your mission area. Your review should be concise; no more than one page (minimum two paragraphs); but must include credible sources that validate the problem you intend to solve.

What to Include in Your Literature Review:

- Problem Context: Briefly describe the broader issue your mission addresses (e.g., lack of digital literacy, healthcare access gaps, agricultural inefficiency).
- Evidence: Reference 5–7 credible sources (reports, studies, articles) that provide statistics or insights supporting the need for your solution.
- Gap Identification: Explain what existing solutions are missing and why your app could fill that gap.

Example (Agriculture Focus): “Smallholder farmers across East Africa face significant post-harvest losses due to limited access to real-time market information [1]. According to the Alliance for a Green Revolution in Africa, nearly 30% of harvests are lost because farmers cannot connect to buyers before produce spoils. Existing digital platforms often require stable internet, which is unavailable in 60% of rural regions [2]. This gap presents an opportunity for an offline-capable mobile solution.”

Problem and Hypothesis Statement (Research-Backed)

Using insights from your literature review, craft a clear problem statement and a testable hypothesis.

Problem Statement :

- [User Group/Persona] is a/an [characteristics] who needs [need] because [research-backed insight].

Example: “Smallholder farmers in rural Rwanda are agricultural producers who need access to offline market pricing because internet connectivity is unavailable in over 60% of farming communities (GSMA, 2023).”

Hypothesis Statement:

- We believe that [solution approach] for [user] will result in [measurable outcome] because [rationale from research].

Example: “We believe that a mobile app that syncs market data when connectivity is available will reduce post-harvest losses by 25% within one harvest season, as supported by pilot data from similar interventions in Kenya (FAO, 2022).”

User Interviews & Empathy Maps

You will conduct at least 10 interviews with potential users from your target audience, with each member in the team conducting at least 2 interviews. Surveys are not accepted; you must conduct live (by talking to your users in-person or via an online meeting/call), interactive interviews to gather qualitative insights. After the interviews, you will create per-user empathy maps and aggregated empathy maps for your target audience.

Interviews allow you to observe emotions, ask follow-up questions, and uncover implicit needs that surveys cannot capture.

Interview Process – Four Steps

Step 1: Recruiting Subjects

Determine your interview goals. Create a screener survey (age, gender, location, job, etc.) to find suitable participants. Invite subjects and schedule time. You may recruit through classmates, friends, family, online channels, hallway recruitment, or existing users. Below is a sample recruitment email, come up with something similar for your team.

Hello **[Participant Name]**,

I am currently working on a project for a certificate program where I need to conduct interviews about **[explain the project]**.

Your participation in the research study would be very valuable to help us **[summary of project goals]**. Details of the study are shared below.

- Location: **[Address and link to map]**
- Study dates: **[Date range]**
- Time: **[Time range]**
- Session length: **[Estimated length of time for each interview]**

With your permission, we will record each interview to help us decide how to improve our product. Recordings will only be shared with project team members for the scope of this project. A consent form is attached to this email that you need to sign and send back.

If you have any questions, please contact me through email at **[your email address]**.

If you are interested in being interviewed, please let me know when you are available between **[date range of study]** and the hours of **[time range of study]**.

Sincerely,

[Your Name]

Step 2: Preparing for the Interviews

Sample interview Script open-ended interview questions. Collect supplies such as a phone, recording app, and notebook. Research your subjects lightly and practice your interviewing skills. Find example scripts below. Your teach will have a document like one of the below for each person you interview. The questions will be the same, the responses is what will differ for each user.

- [Example for food delivery app](#)(By Google)
- [Example for apps scheduling dog walkers](#)(By Google)

Step 3: Conducting the Interviews

Make subjects relaxed by building rapport, thanking them, getting consent to record, and explaining that there are no right or wrong answers. Ask open-ended questions using “why” and “what,” and avoid yes/no questions. Ask follow-up questions based on their responses. Take detailed notes, highlighting compelling quotes and documenting observations such as mood, expressions, and body language. Wrap up the interview by giving the subjects the chance to share their final thoughts and thank them for their time. **Include links to the interviews where users gave consent that it should be recorded.**

Avoiding Bias in User Research: When conducting interviews, be aware of and avoid common biases such as confirmation bias (looking only for evidence that supports your hypothesis), false consensus bias (assuming others think like you), recency/primacy bias (remembering only the first or last thing said), implicit/unconscious bias (unconscious stereotypes affecting whom you interview), and the sunk cost fallacy (sticking with an idea because you’ve already invested time). To prevent bias, ask open-ended questions, interview a diverse set of users, take detailed notes or recordings, use neutral language, avoid leading questions, and space out interviews without rushing.

Step 4: Visualizing and Learning

Draw empathy maps for each interviewed user. Identify user pain points and refine or create user personas.

What is an Empathy Map?

An empathy map is a chart that explains everything you've learned about a user. It has four quadrants: SAYS (what the user says out loud), THINKS (what the user thinks but may not say), DOES (what the user does or how they behave), and FEELS (the user's emotional state).

Create one map per interviewed user, then merge similar users into aggregated maps to form user segments, such as "Positive," "Negative," or "Confused" users.

One-User Empathy Maps



- Imagine you're reviewing an app that promises food delivery in 30 minutes
- Use quotes whenever possible

Aggregated Empathy Maps



- Merge the maps of similar users to form **user segments**
- One aggregated map for each user segment
 - "Positive" vs. "Negative" vs. "Confused" subjects

User Personas

Based on your empathy maps and interview insights, create detailed user persona(s) that represent your primary target user(s). A persona is a fictional but realistic representation of a user segment, including:

About
Evelyn is an account executive from Los Angeles, California. She's a recent college graduate who found a rewarding career in the nonprofit sector. She is actively involved in her community, considers herself a foodie, and loves going to the park with her dog "Bella."

Goals

- Spend more time outdoors
- Make time for friends
- Volunteer with local community organizations
- Practice mindfulness

Frustrations

- Limited parks within walking distance
- Los Angeles traffic
- Expensive dining options
- Did I mention, LA traffic ;)

Quote
“Getting back to nature helps me recharge after a long week.”

Demographics

- Los Angeles, CA, USA
- 24 Years Old
- Single

Social Profile

Twitter, Instagram, Facebook, LinkedIn

Favorite Brands

Apple, patagonia, REI, Target

In UX design, personas are fictional representations of a target user group. The primary purpose of a persona is to provide a clear picture of the group's demographics, as well as their goals, needs, behaviors and pain points.

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Example quote to persona: “John, 22, Computer Science student. Goals: Wake up on time for morning classes. Frustrations: Stays up late gaming, ignores alarms. Habit: Uses phone in bed. Quote: 'I just need something that forces me to stop scrolling and sleep.'”

Personas help keep your design user-centered and guide feature decisions.

Competitor Analysis

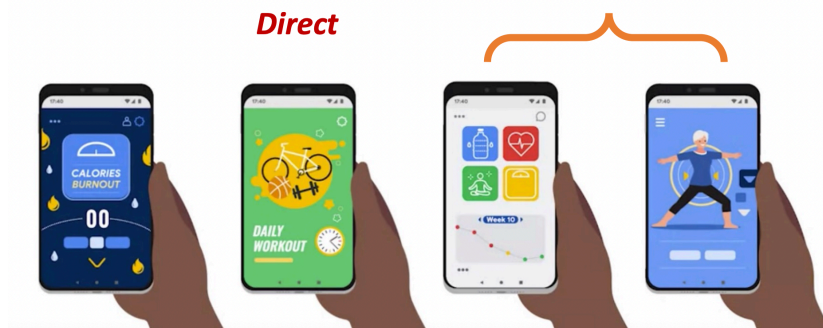
Identify direct competitors, which are products with similar offerings and the same target audience. Also, identify indirect competitors, which may have similar offerings but a different audience, or different offerings but the same audience. For example, for a weight-loss app targeting people in their 20s, a direct competitor would be a health/wellness app for people in their 20s, while an indirect competitor might be a weight-loss app for people in their 60s.

Example
(Weight-loss
App for 20's)

Indirect

- Health/wellness apps targeting 20's
- Weight-loss apps for 60's

Direct



User Journey Mapping

Create a user journey map for an existing solution(a direct competitor) related to your app idea. A user journey map illustrates the steps a user takes to achieve a goal, along with their feelings, pain points, and improvement opportunities(what you will include in your app to make it better than your competitor's).



Persona: Anika

Goal: A fast and easy way to place and pick up group orders

Example (CoffeeHouse App)

ACTION 2.a	Collect orders	Go to Coffeehouse	Submit group order	Wait for order completion	Pick up order
2.b TASK LIST	Tasks A. Collect orders from coworkers B. Collect payment from coworkers	Tasks A. Go to Coffeehouse B. Wait in line	Tasks A. Relay order to barista B. Double-check order for accuracy C. Initiate checkout	Tasks A. Gather any extra items (napkins, coffee sleeves, etc.)	Tasks A. Pick up order B. Check that order is correct
2.c FEELING ADJECTIVE	Excited to connect to coworkers Worried about making order errors	Anxious about getting back to work in time	Stressed about entering each order one by one	Anxious about time	Relieved that order is ready Hopeful that everyone's orders are correct
2.d IMPROVEMENT OPPORTUNITIES	Offer a way to easily collect multiple orders	Create an app for advance ordering	Offer a way to easily collect multiple orders	Create an app that offers order status updates	

Unique Value Proposition

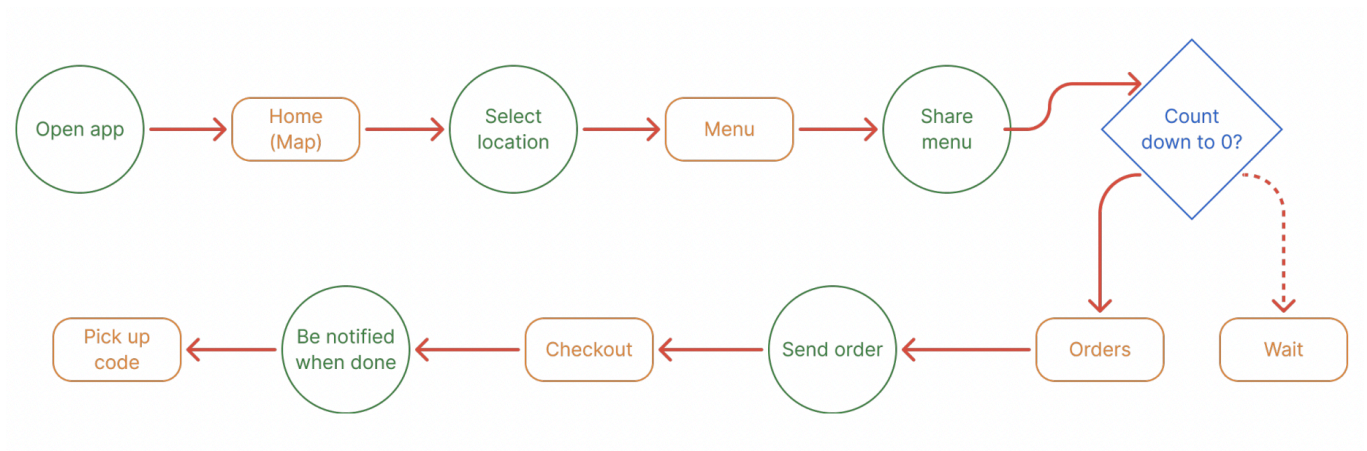
Based on your problem statement and findings from preliminary research, user interviews, and competitor analysis, articulate your app's Unique Value Proposition (UVP). This is a clear statement that describes the benefit of your product, how it solves users' needs, and what distinguishes it from the competition.

Your UVP should answer: Why should users choose your app over others? What unique perspective or solution does it offer?

Example: "Our app provides personalized, gamified sleep scheduling that turns bedtime into a rewarding routine, unlike generic alarm apps."

User Flows

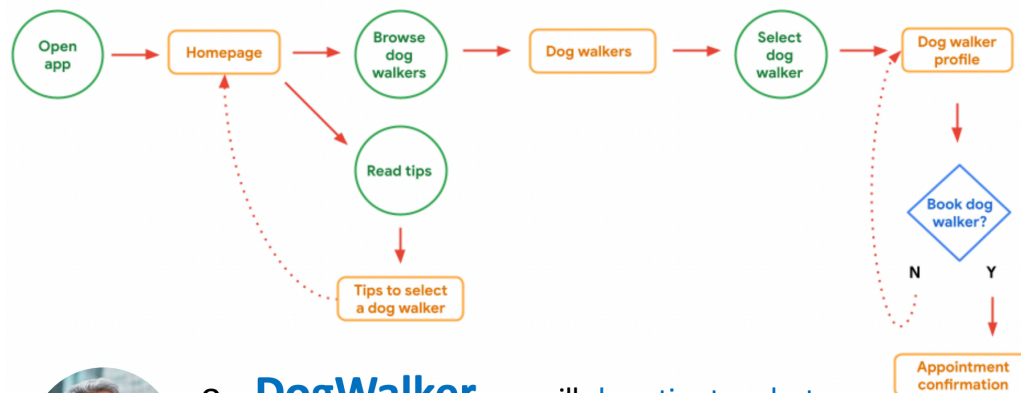
Design the key user flows for your app. A user flow is a diagram that maps out the steps a user takes to complete a task within your app. It includes screens, decisions, actions, and possible outcomes (happy path and edge cases). Create a flow for each core task (e.g., signing up, booking a service, completing an order). Use simple shapes and arrows to illustrate the path. This will serve as the blueprint for your prototype



Storyboards

Storyboards are a series of panels or frames that visually describes and explores a user's experience with your app's solution. Below is a sample to refer to and create one for your app.

Exercise: DogWalker Storyboards

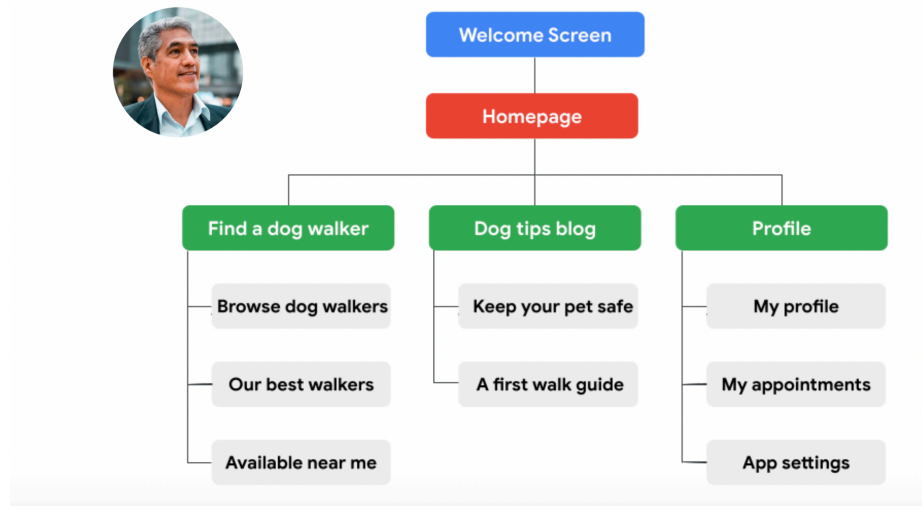


Our **DogWalker** app will show tips to select a dog walker which will affect non-tech-savvy users by helping users hire the right dog walkers. We will measure effectiveness by tracking the booking rate in the app.

Information Architecture (IA)

Organize your app's content and structure to help users navigate intuitively. Define the main sections, screens, and navigation patterns. Consider principles like the Choice Principle (fewer, well-organized options) and the Focused Navigation Principle (logical menu design). Create a sitemap or navigation diagram that shows the hierarchy of screens and how they connect.

Example: DogWalker



Figma Prototype Design

Translate your user flows and information architecture into an interactive, high-fidelity prototype using Figma. Your prototype should include:

- High-Fidelity Mockups: Polished screens with consistent typography, color (using the 60-30-10 rule), icons, and imagery.
- Interactive Elements: Clickable buttons, navigational links, transitions, and simulated user interactions.
- Key Screens: Home, main task flows, profile, settings, and any unique features.

Your prototype should reflect the insights from your interaction with users via interviews, address user pain points, and demonstrate your unique value proposition.